Specifications

Next Generation Security Information and Event Management (NG SIEM)

Sl. No	Specifications
1	SIEM solution must be dedicated on premise solution. The Bidder should propose the necessary hardware along with any additional software licenses (Database, Virtualization) as part of the proposal if required.
2	Bidder should ensure OEM support for the solution and the escalation matrix in regard to 24*7 support is also required to be submitted.
3	The platform should be offered along with External Attack Surface Monitoring capability, either natively or as an additional modules, if offered as a SaaS module (the Attack surface monitoring tool should have the capability to send the information to the SIEM solution placed in on-premise/Airgap Network)
3.a	Should conduct automated and continuous scanning of the Organization's publicly exposed hosts and networks
4	The solution must provide the ability to encrypt communications on the network between SIEM components and SIEM.
5	The solution must ensure all distributed system components continue to operate when few parts of the NG-SOC solution fails or loses connectivity (i.e. Management engine goes off-line all separate collectors continue to capture logs).
6	The solution should be proposed along with hardware &software based with hardened OS or big data- based platform or equivalent technology. Clearly elaborate the components constituting SIEM.
7	High-Level Diagram to be submitted for SIEM solution ensuring no data loss and optimal bandwidth utilization in WAN and LAN.
8	The solution should demonstrate 'ease of use'. Ease of use is critical to the successful deployment and on-going use of the solution.
9	The solution should allow a wizard-based interface for rule creation. The solution should support logical operations and nested rules for creation of complex rules.
10	Collection, Co-relation and Console layer should be physically or logical separate.
11	The solution should support log collection, correlation and alerts for the number of devices mentioned in scope. The proposed solution should be able to pull the logs through JDBC / ODBC connectors out of the box.
12	The SIEM should support a Sustained EPS of 6000 EPS at all layers from day one. It should be able to handle a burst of 2 times of the sustained EPS in real time at any given point in time without any drop or queuing of events. Solution should have 6000 EPS capacity across all layers of the solution i.e. log collection, core NetFlow logs, correlation and management respectively. Architecture, licensing and hardware & software sizing must be designed and planned accordingly. There should be no limitation on number of devices to be integrated.

13	Raw and normalized Logs should be handled and stored in tamper proof way across SIEM solution.
14	The solution must provide a complete audit trail and accountability during the incident handling for forensic investigations. The system should have ability to perform event forensics to determine what really happened before, during, and after the event.
15	The solution should have customizable parsers to accept and process unknown log formats. Raw logs should be visible to the user in one single click. Raw log data from the solution shall be made downloadable without the need of OEM dependent tools.
16	The solution must provide capabilities for time stamping, efficient storage and compression (minimum 50%) of collected data.
17	The solution must support/normalize event time stamps across multiple time zones.
18	The system should provide the ability to write a custom parser or filter for an unknown new event and a new log source.
19	The solution shall allow bandwidth management, rate limiting, at the log collector level.
20	SIEM OEM should have their own agents to natively collect logs. No Open source agents should be used to collect logs.
21	Log Search Interface: The proposed solution must provide a simple, intuitive search interface using following search methodologies: a) Search Drilldown b) Search Patterns c) Search Operators d) Regular Expression e) Flow-based Searches f) Search Criteria g) Search Time Range h) Search Results View i) Search Export j) Search Combinations
22	The solution must have an automated backup/archival/recovery process. Please mention expected downtime for recovery process.
23	The solution must provide near-real-time analysis of events. Mention lag, if any, between the actual event and its reporting with analysis.
24	The solution should provide the ability to aggregate and analyze events based on a user specified filter. Give the list of in-built filters (IP addresses, usernames, MAC address, log source, correlation rules, user defined, etc.) available. Also explain the ease of use of filters.
25	Log Management: The proposed solution must provide a log management solution and must retain minimum of latest 3 (Three) Months of Online logs (RAW + Normalized) and rest of 9 (Nine) Months. Minimum 12 Months log must be available at any point of time. However, if Organization wish to store online/offline logs for more duration, it shall be possible without any cost implication

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26	Universal Log Analysis: The proposed solution must contain system content that can be used for cyber-security, compliance, application and IT & OT operations monitoring and must support additional content specific to regulations like ISO 27001 ,IT-Act etc..
27	The proposed SIEM solution should highlight the number of parsed/unparsed logs per hour for each sub organisations, i.e .SCADA/EMS, URTDSM, REMC, ABT OA & MIS, Islanding Scheme etc
28	Log Data Integrity: The proposed solution must provide audit trail of all the administration activities such as login, logouts, new user creations, etc. and alerting mechanisms in case of any access/change.
29	Search Performance – Structured Data: The proposed solution search performance must be capable of searching through millions of structured (indexed) events
30	Search Performance–Unstructured Data: The proposed solution search performance must be capable of searching through millions of unstructured(raw)log messages
31	Saved Search Filters: The proposed solution must provide a simple, intuitive way of allowing users to save search filters for later use and to be shared with other authorized users.
32	Historical Analysis: The proposed solution must be capable of processing and storing large volumes of historical log data that can be restored and analyzed for forensic investigation purposes.
33	The proposed SIEM solution should highlight the number of logs from each individual log source.
34	Retention Policies: The proposed solution must provide the ability to define retention policy for group/ individual devices. The system should have the capability to configure individual device based retention policies.
35	Retention Enforcement: The proposed solution must enforce data retention policies automatically without necessitating manual data disposition or clean –up efforts.
36	Retention Policy Suspension: The proposed solution must provide the ability to suspend the retention policy manually and allow administrators to increase the retention period dynamically for the purpose of evidence preservation in the event of pending litigation.
37	The proposed solution should have the capability to upload files offline and create visualizations for the same.
39	The proposed SOLUTION must provide the following capabilities:
	a. Incident review framework to facilitate incident tracking, investigation, pivoting and closure. b. Threat intelligence framework that automatically collect, aggregate indicators of compromise from threat feeds.
40	Solution should support the collection application log data from custom / in-house developed web applications, with or without explicit custom parser development.
41	The solution should be able to dynamically enrich data based on set of parameters (if certain parameters are met, then only enrich the log)
42	The proposed SOLUTION must be able to provide the capability to annotate events, modify status, and build a chronological timeline for the incident before and after a triggered event.

43	The solution should be able to collect and process raw logs in real- time from any IP Device including Networking devices (router/switches/voice gateways), Security devices (IDS/IPS, AV, DAM , Firewall/DB Security solutions), Operating systems (Windows (all flavors), Unix, LINUX (all flavors), AIX etc), Mainframe(z/196), Virtualization platforms, Databases (Oracle, SQL, DB2etc.),Storage systems, and Enterprise Management systems etc.
44	The solution should be able to conduct agent less collection of logs except for those which cannot publish native audit logs.
45	The solution should support search within the results of a forensic search
46	Distributed Event Processing: The proposed solution must collect logs in a distributed manner, offloading the processing requirements of the log management system for tasks such as filtering, aggregation, compression and encryption.
47	Custom Collection API: The proposed solution must have a software tool to allow customers to create integration with unsupported legacy or internally developed event sources. The software tool must allow customers to integrate with Syslog, log files, databases etc. and support the ability to parse multi -line log files.
48	Categorized Event Data: The proposed solution must categorize log data into an easy-to-understand humanly-readable format that does not require knowledge of OEM-specific event IDs to conduct investigation, define new correlation rules, and/or create new reports/dashboards.
49	Secure Transport: The proposed solution must provide encrypted transmission of log data from device to SIEM system.
50	Reliable Transport: Log Transmission should use reliable TCP protocol that will ensure retransmission in the event of protocol failure to ensure that no log data is lost in transit.
51	Collection Health Monitoring: Any failures of the event collection infrastructure must be detected immediately and operations personnel must be notified via various communication mediums such as e -mail, ticket etc. Health monitoring must include the ability to validate that original event sources are still sending events.
52	Event Filtering: The proposed solution must provide inline options to reduce event data at the source by filtering out unnecessary event data.
53	Event Aggregation: Aggregation must be flexible in which normalized fields can be aggregated and provide the ability to aggregate in batches or time windows.
54	Caching & Batching: The proposed solution must support local caching and batching at collection level in case of connectivity failures.
55	Compression: Proposed solution should allow compression to conserve bandwidth.
56	Raw Event Data: proposed solution must support the option of collecting raw event data using Syslog, FTP, SCP, SNMP protocols, and any other protocol required for collection of logs etc. This ensures original audit quality data is available for forensics.
57	Windows Event Logs: The proposed solution must be able to integrate with a Windows Domain in an agent-less fashion and collect the event logs from multiple systems without requiring any agents to be installed on the end devices.

58	Time Correction: The proposed solution must be capable of collecting event time for systems along with collection time and alerting time. This allows integrity for forensic analysis to determine the original time of the event source and what the system time was for each system component processing the event.
59	The solution should be able to conduct agentless collection of logs except for those which cannot publish native audit logs. System should not leverage any open source agents like OSquery, Wazuh, fluentd etc & instead the agents should be from the same OEM..
60	The solution should have native geo-location feature.
61	The solution proposed should collect and analyse audit trails logs and Netflow information (all types of logs – ODBC, Syslog, etc.) to detect malicious or abnormal events and raise the alerts for any suspicious events that may lead to security breach in the scoped environment.
62	The solution should have the ability for Raw Log Backup.
63	Support for operational technology (OT) and Internet of Things (IoT) technologies and environments (e.g., ICS/SCADA).
64	The EPS burst should be processed in real time without dropping or queuing to ensure real time analysis of threat.
65	A template for each application should be made consisting of format of log and type (IIS, https, transaction log, login/logout audit log for each application etc..) , future applications should be be able to send log data to SIEM through log collection APIs.
	Centralized Incident Management: The proposed solution must provide an interface to view the incidents and alerts generated. The solution should have integrated Incident review frameworks such as MITRE ATT&CK and Attack Kill Chain integrated.
66	Correlation Rules: The proposed solution must provide many correlations rules out-of-the-box to automate the incident detection and workflow process.
67	Cross-Device Correlation: The proposed solution must be capable of correlating activity across multiple devices.
68	The solution should have intelligence to minimize false positives alerts, correlate and deliver accurate alerts.
69	The solution must support the ability to correlate against vulnerability assessment tool.
70	The solution must monitor and alert when there is a disruption in log collection from a device. In other words, if logs are not seen from a server in five minutes then generate an alert.
71	The solution must support correlated incidents for applications, databases, servers, networks etc. based on feed from other solutions like PAM, WAF,VAPT,NBAD,TIP, Threat hunting Centre and UEBA.
72	The solution must provide many correlations rules out-of-the-box. Again, option of creating/configuring new rules must be available. Please provide the complete list/count of rules which are available out of the box from the system
73	Solution should be able to perform the following correlations (but not limited to)based on analysis rules mapped to various threat categories and provided with criticality information. The various threat categories to be covered include:

	1) Vulnerability based 2) Statistical based 3) Historical based 4) Heuristics based 5) Behaviour based on source entity, applications etc. 6) Information Leak 7) UnauthorizedAccess 8) Denial of Service 9) Service Unavailable 10) Phishing attack 11) Pattern based rules 12) Profiling 13) Whitelist/Blacklist/Reference List
74	Solution should support at least rule based, non-rule based, vulnerability based and statistical based correlation.
75	The solution should have out-of-box rules for popular IDS, firewalls, antivirus, operating systems, etc. Documentation of the correlation rules should also be provided.
76	The solution should have intelligence to extract Information from leading global intelligence sources, threat intelligence platform and use it for valid correlation.
77	The solution should be a unified platform with SIEM, SOAR, UEBA accessible from same dashboard.
78	The system should provide the capability for identify zero-day threats on the network.
79	The system should have ability to perform multiple event correlation to process all time and transaction- based events to provide actionable data and incident awareness.
80	The system should provide real-time as well as historical correlation of events. This includes the techniques used for correlation of different events across different monitored devices. Describe the process for handling both real-time events and historical events.
81	The system should provide adequate categorization and prioritization of the collected and aggregated events from the monitored log sources. This entails a deep understanding of the event types and criticality associated with the events for the supported log sources. The categorization may by be HIGH, MEDIUM, LOW or color coding.
82	The system/solution should have the ability to correlate all the fields in a log.
83	The solution must leverage both Supervised and Un-supervised Machine learning techniques without signatures
84	Events should not be dropped till it exceeds the scalable EPS limits.
85	The solution must provide a mechanism to capture all relevant aspects of a security incident in a single logical view. This view should include relevant events, network activity data, correlated alerts, vulnerability data, etc.
86	Future Proofing: The proposed solution must provide a level of confidence that reporting will continue to work and not have to be modified if a particular technology, such as a Firewall or IDS product, is replaced with a newer product or OEM. The reports should continue to run and

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	include the new technology into the report criteria automatically.
87	The solution should visualise individual log source wise dashboards.
88	Custom Dashboards: The proposed solution must provide the framework to create custom visual displays for any business group.
89	Dashboard Drill-Down: The proposed solution must provide the ability to allow analysts to drill -down from graphical dashboards to the underlying event data.
90	Content Management: The proposed solution must provide the ability to synchronize its resource contents (i.e. rules, dashboards, reports, filters etc) automatically across multiple instances of the product, to support multi instance/high-event-rate deployments.
91	Statistical Correlation: The proposed solution must be capable of keeping a statistical baseline of "normal" monitored activity. This includes attacker, target, ports, protocols.
92	Correlation Flexibility: solution must be capable of running cross device correlation, real time correlation at the same time.
93	The proposed SIEM solution should have the ability to customise login banners
94	The solution should showcase the real time logs coming in from each and every source and should have the ability to filter out specific sources and specific keyword based filters in realtime data ingestion.
95	Dynamic / Static Lists: The proposed solution must allow users to define either whitelist or blacklists that can be used as inclusion or exemption during the correlation process. Additionally, the correlation engine should utilize dynamic lists to provide important information such as shared user monitoring, attack history and privileged system access. Product must support import capability to create/update monitoring list which can be dynamically add/ remove values without manual intervention
96	Correlation Performance: The proposed solution must be capable of efficiently presenting categorized data to the correlation engine to allow real -time detection and response.
97	Advanced Rule Chains: The system must provide the ability to allow rules to be triggered in a series, matching various correlation activity before an alert is generated.
98	Alert Thresholds: The proposed solution must provide the ability to aggregate and suppress alerting with granular options and use conditional logic to determine if an alert should be generated
99	Re-Usable Content: The solution must allow users to create objects such as filters or search queries that are reusable throughout the system.
100	Content Editor: The proposed solution must provide a common interface to create or modify resources within the system. All aspects of this editor must apply to the development of rules, reports, dashboards and any other resource that will be created in the system.
101	The solution should be able to collect and store data from various devices as text/csv files and use it for analysis. All the latest updates and patches for the solution should be updated to the NG-SOC without any additional cost The solution should allow creation of custom reporting dashboards from forensics search

	conducted on the system
102	The solution must provide real time alerting based on observed security threats. The critical alerts should be transmitted using multiple protocols and mechanisms such as email, SMS, voice call etc. based on agreed policies.
103	Solution must be capable of monitoring attack/incident history against critical assets or by particular users.
104	The solution should have option to assign priority against the alerts to allow prioritization based on multiple configurable characteristics such as asset type, protocol, application, etc.
105	The solution must display traffic profiles in terms of packet rate/ traffic volume/ protocol.
106	Real-Time Alerts: The proposed solution must be capable of generating alerts based on filter pattern matches for operational health monitoring.
107	Threshold Alerts: In addition to real -time alerts, the system must provide threshold alerts, configured from saved search queries.
108	Alert Filters: The proposed solution must provide pre - defined alerts and provide the ability to re-use pre-defined filters and own created filters as alert criteria.
109	Alert Delivery: The proposed solution must provide options of how alerts are delivered to operations or security personnel. At a minimum the options must include reporting to the web console, send an email, generate an SNMP trap to an external management system, and send alert on mobiles. The solution must be capable of doing all these concurrently for each alert.
110	The solution must provide a mechanism to capture all relevant aspects of a security incident in a single logical view. This view should include relevant events, network activity data, correlated alerts, vulnerability data, etc.
	In case the connectivity with the SIEM management system is lost, the collector should be able to store the data in its own repository. The retention, deletion, synchronization with SIEM database should be automatic but it should be possible to control the same manually.
111	The centralized web based/console user interface should drill down on reports and incident alerts on real time basis.
112	The solution must provide reporting engine for out-of- box reports, customized reports, ability to schedule reports, compliance reports, historical trend reports with the following options: 1. Detailed reports of non-compliant activities and policy violations in the network. 2. Reports should be available in pdf and csv format. 3. Information about threat response and mitigation measures carried out to prevent attacks. 4. The solution must provide reporting engine for out-of-box reports, customized reports, ability to schedule reports, compliance reports, etc.

	5. The solution should provide out of box templates for reports on ISO, PCI, SOX and other standards. 6. The solution must support direct drill-down from the reports and charts. 7. The system should allow scheduling reports.
113	The solution should allow users to initiate and track alert related mitigation action items. The portal should allow reports to be generated on pending mitigation activities.
114	The proposed SIEM solution should highlight the number of parsed/unparsed logs per hour for each sub organisations.
115	Reports should be possible to be scheduled and mailed across to the requisite person.
116	All out of box content should be made available for use as and when published by the OEM
117	The solution should provide out of box templates for reports on ISO 27001 standards at no additional cost.
118	Pre-Defined Reports: The proposed solution must provide pre- defined, out-of-the-box reports for Operations, Security and Compliance that can easily be modified.
119	Compliance Reports: Solution should provide compliance auditing, alerting and reporting for governances for ISO 27001.
120	Customized Reports: The proposed solution must provide the ability for customers to create their own reports with report templates
121	Report Export: The proposed solution reporting function must be capable of exporting reports in various formats. At a minimum, the report formats should be, .xls, .csv,.pdf etc.
122	Report Scheduling: The proposed solution must provide the ability for customers to schedule and email reports to run hourly, daily, weekly or monthly as an attachment. There must be numerous output formats and delivery options for scheduled reports.
123	Run-Time Report Options: The proposed solution reporting engine must provide the ability to filter, highlight, and modify various report functions at runtime.
124	The solution should provide an integrated case management system which should ensure independent investigation eliminating the risk of possible intervention of administrator.
	Dashboard
125	The proposed SIEM solution should highlight the number of parsed/unparsed logs per hour for each sub organisations/tenants
126	The centralized dashboard to monitor the alerts and events from all devices of Organization at its locations and from the tools provided as a Part of NG-SOC solution.

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127	The solution should provide customizable management console/dashboard which can be provided to different Teams. Access to the solution should be restricted based on role of that team/user, which should be configurable
128	The solution dashboard should be in the form of a unified portal that can show correlated alerts/ events from multiple disparate sources such as security devices, network devices, enterprise management systems, servers, applications, databases, etc.
129	Events should be presented in a manner that is independent of device specific syntax and easy to understand for all users.
130	The dashboard should show the status of all the tools deployed as part of the SOC, including availability, bandwidth consumed, system resources consumed (including database usage)
131	It should be possible to categorize events while archiving for example, events for network devices, antivirus, servers etc.
132	Customizable Dashboards: The proposed solution should provide dashboards specific to each user and should be user configurable. The dashboards must be capable of displaying multiple daily reports Specific to each users job function.
133	Solution should provide Dashboard not limited to Audit Dashboard, Security Dashboard, Risk Dashboard, Analytics Dashboard, Asset Dashboard, User Activity Monitoring dashboard, User/ Identity Dashboard, Threat Intelligence Dashboard. Etc.
134	The solution should showcase the real time logs coming in from each and every source and should have the ability to filter out specific sources and specific keyword based filters in Real-time data ingestion.
135	Solution should have dashboards to identify and investigate security incidents, reveal insights in your events, accelerate incident investigations, monitor the status of various security domains, and audit the incident investigations.
136	Solution should help to investigate incidents with specific types of intelligence. a. Threat intelligence dashboards use the threat intelligence sources and custom sources that you configure to provide context to your security incidents and identify known malicious actors in your environment. b. User intelligence dashboards allow you to investigate and monitor the activity of users and assets in your environment. c. Web intelligence dashboards help you analyse web traffic in your network and identify notable HTTP categories, user agents, new domains, and long URLs
137	Dashboard Integration: The proposed solution must be accessed through web interface so that display dashboards, queries and reports can be executed and viewed.
138	The dashboard should drill down on events and find the IP addresses and geo-locations from the sources of suspicious or malicious IPs.
139	SIEM solution should be able to map correlation rules/use cases with MITRE ATT&CK Framework and Cyber Kill Chain for tactics and techniques to get better visibility of incidents and shall be a part of the proposed solution

140	The SIEM solution should be integrable with DAM for monitoring any management activity of the solution.
141	NG-SIEM must be integrable with all other proposed solutions part of the RFP
142	The solution should have connectors to support the listed devices / applications, wherever required the Bidder should develop customized connectors for all standard devices at no extra cost
143	The solution must normalize common event fields (i.e. usernames, IP addresses, hostnames, and log source device etc.) from disparate devices across a multi- Bidder network. Solution to provide the ability to normalize and aggregate event fields that are not represented by the out-of-the-box normalized fields.
144	The solution must support information collected from File Integrity / Activity Monitoring (FIM / FAM) Security software and tools.
145	The proposed solution should have the capability to upload files offline and create visualisations for the same.
146	Solution should integrate with IDS, IPS, Firewall etc to consume alert data.
147	The proposed SIEM solution should have the ability to customise login banners
148	In case the connectivity with SIEM management system is lost, the collector should be able to store the data in its own repository. The retention, deletion, synchronization with SIEM database should be Automatic but it should be possible to control the same manually.
149	Alerting: The proposed solution should provide the ability to integrate with enterprise-class network management systems through SNMP.
150	Syslog Forwarding: The proposed solution must be able to receive raw (i.e. unprocessed) event data in the form of syslog messages or text log files, in addition to receive the raw original event data from collectors.
151	Threat Intelligence feed should be a part of the proposed SIEM solution by bidder for identifying new global threats
152	Bidder should provide global threat intelligence which combines internal data with third-party data on threats and vulnerabilities and attach patterns as per of the proposed solution without any additional cost on it.
153	Solution should have an OOTB bidirectional integration with Threat Intel Platform.
154	The system should receive feeds from a threat intelligence repository maintained by the OEM which consists of inputs from various threat sources and security devices across the globe.
155	The system should be capable to consume Threat Intelligence from Third Party sources as well.
	Administration

156	The Solution should provide web-based administration user interface for device management and monitoring.
157	The system should support Network Time Protocol for time synchronization.
158	The solution should be hardened and should be CERT-IN VA approved.
159	In case the connectivity with SIEM management system is lost, the collector should be able to store the data in its own repository.
160	There should be an automated switch over to secondary collector in case of failure on the primary collector. No performance degradation is permissible even in case of collector failure.
161	SIEM Solution should have a common interface/native integration with UEBA, SOAR solution.
162	The monitoring capabilities to ensure that the proposed system is functioning under optimal parameters e.g. CPU/storage etc.
163	Administrative Interface: The proposed solution must provide a web / thick client interface used for administrative tasks including but not limited to initial configuration, updates, patches, backups, content creation, analysis, user management and all other tasks.
164	Administration Dashboard: The proposed solution must provide a single administrative dashboard to analyze the system load, event Flow and storage performance trends.
165	The solution should have the capabilities to fetch data records upto 5 million from day one in a single search
166	Simple System Backup: The proposed solution must provide a simple method for manually backing up the system.
167	Device Discovery: The proposed solution must automatically accept log data from any system that is reporting through network. All log data, once received and indexed should be available for searches, alerts, and reports.
168	System Process Status: The proposed solution must provide administration page that allows viewing underlying system process status and resetting application components without having to restart the entire system. This should be provided through the same Web interface along with all other administrative tasks.
169	SSL Administration: Solution should have Self-signed certificate generation features so that accessing of appliance from client for monitoring and administration purposes can be done in encrypted manner.
170	Administration Audit Trail: The proposed solution must log all administrative access and activities and provide access to the audit logs through the same web interface.
171	Administration Alerting: The proposed solution must provide the ability to alert on system state activity such as low disk space, component failures, high resource utilization, etc. The transport for these alerts must be simple to configure and support SMTP, SNMP, Syslog, and/or direct SIEM integration.

172	The solution should provide intuitive mechanisms for troubleshooting such as proactive notifications, command line utilities etc.
173	The solution should support the automatic update of configuration information with minimal user intervention. For example, security taxonomy updates, Bidder rule updates, device support, etc.
174	Threat Hunting Features:
a.	The solution should have in-built ML based searches to perform threat hunting on 24x7 basis/real time
b.	The solution should give ability to perform open searches with simple and complex queries and enable threat hunting
c.	The solution should give ability to store queries and execute queries on a periodic basis/as per requirement.
d.	The solution must give capability to perform forensics and raw data download without depending on any query language.
e.	The solution must have out of the box machine learning algorithms for Threat hunting which will execute queries on 24x7 basis
f.	All the Event, Alerts and other information pertaining to Organization's NG-SOC must remain within Organization premises only. Any information moving out of Organization premises shall be reviewed and approved by Organization on need basis.
g.	Solution should support all four categories of threat hunting including Network Threat Hunting, User Behaviour Anomaly Hunting, End Point Threat Hunting, and Application Threat Hunting.
h.	Network Threat hunting should utilize existing logs from security controls such as firewalls ,IPS devices , Web Security Appliance(Proxy), Anti APT solutions to detect targeted attacks. Should support Real Time Anomaly Detection with single actor, Interactions among actors, Learning with higher weight on recent actions. Solution should support pre-defined alerts and support email or text notifications, along with functionality to email comprehensive periodic reports. Solution should provide the built-In alert workflow.
175	The solution should have out of the box capability to integrate with CERT-In CMTX threat feeds & NCIIPC feeds.
176	Should be able to collect logs from all the devices.
177	Log collector should also act as a connector for SOAR to take remediation actions and access devices which are behind a firewall.
178	The log collector should be able to compress the logs it collects to at least 50% and transmit the logs in an encrypted format.
179	The logs collector should have the capability to store logs in case connectivity fails. The collectors should be sized to store at least 5 day's worth of data.
180	The log collectors shall use TLS encryption to forward data.

181	The log collector shall have inbuilt JDBC/ODBC Connectors for collecting logs from Databases.
182	The log collectors should be capable to be deployed in Physical & Virtual Server Environments
183	The log collectors should have the ability to collect flows if required.
184	The log collectors should have the capability to be deployed in HA.
185	The log collector should have the capability to generate network metadata upto layer 7 from the raw network traffic.

Security Orchestration Automation and Response (SOAR)

Sl. No	Specifications
1	The solution must be a fully on-premise solution deployed in house. The OEM to recommend the sizing for the hardware/VM for the proposed solution
2	SOAR should be able to integrate bi-directionally with SIEM solution being proposed from day 1.
3	SOAR platform must have out-of-box or ready integration with the various technologies used in Organization environment to consume alert data, perform investigative and remediation actions.
4	The solution must be able to support creation of incidents via API, Web URL, SIEM, Ticketing system, manually etc. SOAR platform must have out-of-box or ready integration with the various technologies used in Organization environment (attached as Annexure-II) to consume alert data, perform investigative and remediation actions
5	The solution must retain all incident related files not limited to malware specimens, logs, artifacts/screenshots for minimum of Twelve (12) months. However, if Organization desire to keep for further period same shall be retained without any additional cost.
6	The solution must have capability to design workflow to provide fully automated action for the detected incident.
7	The solution must have the capability to notify user based on detected/ identified incidents.
8	Solution shall have the capability of providing independent threat intelligence for local and external threats.
9	Solution should support 10 security admin analysts and unlimited read only users from day one and provide Role Based Access Control (RBAC) to differentiate between analysts and administrators for restricting access to investigations, Jobs, Scripts, playbooks and admin tasks with support for storing upto million indicators in database from day one.
10	Solution should support deployment for access remote networks which are behind the firewall or isolated from Internet

11	Solution should be able to integrate with security devices like Firewall, IDS/IPS, endpoint Security solution, APT solution, WAF,PIM etc from day one and the other proposed NG-SOC tools.
12	The solution should provide for Threat Intelligence and Threat hunting capability via integration with the proposed TIP and Threat hunting platform.
13	The solution must be web based without the need for installing an additional client software for administration and routine day to day usage requirements.
14	All the Event, Alerts and other information pertaining to SLDC'NG-SOC must remain within SLDC premises only.
15	Any information moving out of SLDC premises shall be reviewed and approved by SLDC on need basis.
16	Solution should support storing all IOCs/indicators in database for 180 days from day 1.
17	Solution should support backup / restore and should be provisioned for HA mode.
18	Solution should support Multi Factor Authentication
19	The system should support a graphic UI for creations of playbooks
19.2	The solution should have live visualization of logs received from each source
19.3	The system should support a graphic UI for creations of playbooks. The flow of playbooks should be in form of drag and drop.
19.4	The SOAR capability should include running of custom scripts such as but not limited to bash, python, ruby inside a playbook
19.5	The solution should offer a dedicated SOAR test lab workspace, empowering users to experiment, validate, and enhance automated security workflows in a controlled testing environment
20	Solution should support integration with min 100 third party OEM products including but not limited to the following technologies. > Forensic tools > IT tools (AD, ISE, NOC tools) > Specify all products IT e.g. (AD, SAML) Communication tools (e.g. Emails, SMS) SIEM tools. > Endpoint Security Solution > Network Security Solution > Threat Intelligence. > Dynamic malware analysis
21	Solution should support adding of new product integrations and custom integrations without any additional cost to SOC.
22	Solution should integrate with Incident management tools and ticketing system. Solution should support Integrations with external ticketing platforms.
23	The solution should integrate with partner products using any of the standard protocols and interfaces including REST API, SOAP, SSH/CLI interface, and custom APIs

24	The solution must provide the capability to integrate multiple threat intelligence feeds from various providers to enrich incident artefacts.
25	The solution should provide a simple, comprehensive, fully automated approach to detect and stop the threats that matter, for on premise deployments from internal & external attacks on Organization IT and OT system.
26	The solution should support both human and machine-based automation for various tasks related to security investigations
27	Solution should support addition of automation scripts to existing integration
28	For secure operations, the solution must run various scripts, commands, application functions, playbooks etc without the need of running with elevated privileges on a host OS.
29	Solution should use playbooks/runbooks with a visual editor/canvas which supports visual creation of playbooks without the need to code by native integration to third party tools and processes.
30	Solution should auto remediate the problem without causing a huge impact to the organization. Some of the examples such remediation could be: · Push policies to prevent an external IP · Isolate an internal desktop/Server · Disabling user accounts used for malicious purposes
31	Solution should be configured with the used cases with automation for response to the minimum basic threats like: • Blacklisted IP Communication • Possible Penetration Testing Activity • Connection to Known Malicious Actor in Published Host List • DDOS Attack • Vulnerability scan detection • Phishing detection • Brute force attack • Malware /threat activity monitoring • Ransomware • Buffer Overflow attacks • Port & vulnerability Scans • Worm/virus outbreak • File access failures • Unauthorized server/service restarts • Unauthorized changes to firewall rules
32	Solution should have min 50 built in reusable playbooks for well-known Incident types (Phishing, Malware, IOC Hunt).
33	Solution should allow creating new playbooks to map out the CIRT processes. Provision for building custom playbooks should be factored within the solution.
34	Solution should support re-use of playbooks in bigger playbooks
35	Solution Should allow creation of Manual Tasks, Automated Tasks and Conditional Tasks in Playbooks

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36	Solution should allow a single playbook to have Automated and Manual Tasks within the same playbook
37	Solution should allow a complete playbook to be run automatically or manually and list out any exceptions
38	Solution must support step by step debugging of the running playbooks with provision of starting from where it stopped on error
39	Solution should record all manual and automated entries during execution of a playbook
40	Solution should allow addition of adhoc tasks within a playbook
41	Solution must support provision to pass parameters to upstream/downstream task within a playbook.
42	The solution must have an integrated versioning mechanism to save and maintain multiple versions for the playbooks.
43	The solution should allow for viewing version history for all or selected playbook and provide option for restoring to an older version
44	Solution should be able to do incident analysis on the data received and should be an input for subsequent playbooks. The collected data can be used for incident analysis, and also as input for subsequent playbook tasks.
45	Solution should support updates for Playbooks, Integrations and should specify the procedure to update each of them. The solution should not limit the number of SOAR playbooks to be created.
46	The solution should be able to find related incidents from historical data based on assets like IPs or user involved in incident
47	The system should support parsing all the SIEM message fields, including but not limited to: creation time; update time; source/dest IP; source country; category; system; rule-name; severity; dest IP
48	The system should support automatic reporting back to ticketing solution for example for closing cases state. These actions will be added to the audit trail.
49	The solution should be able to consume security alerts/incidents from SIEM or directly from any other IT security solutions.
50	Solution should support email or text notifications, along with functionality to email comprehensive periodic reports and dashboards.
51	Solution should provide content for threat descriptions as well as remediation advice.
52	Solution should provide necessary integration with the IT/cybersecurity systems for keeping the forensics artefacts from the integrated sources of the incident before taking remedial actions.
53	SOAR platform maybe positioned as end to end incident management, Incident response, incident remediation, investigation platform and single evidence repository by the bidder and OEM. Platform should be capable to provide detailed post incident documentation about all the actions taken, root cause, controls implemented etc.

54	Solution should provide an integrated incident management platform for Security and IR team
55	Solution should support assigning of incident to a User or a group
56	Solution should maintain SLA for incident
57	The solution must have a provision to remove duplicate incidents and merge all duplicate ones in a single incident automatically and manually.
58	Solution should support highlighting of active incidents to quickly identify and access them.
59	Solution should document all artefacts related to an incident
60	Solution should support searching of Data/artefacts associated with historical incidents
61	Solution should support visual mapping of an incident, its elements and correlated investigation entities, and the progression path of the incident.
62	Solution should support external users to contribute to an incident via email, message etc.
63	Solution should highlight if any external products are required for Collaboration. It should provide an exhaustive list of such products currently supported.
64	Solutions should support sharing of knowledge between users using its own platform
65	Solution should provide an interface to drive High priority Security Incidents by Security teams and provide access and visibility of this incident to management, legal etc. without any additional cost to licenses
66	Solution should support key entities/IOCs for every incidents which can be auto extracted and presented in a graphical/ tabular form for analysts to view relationship between key entities for an incident.
67	System should allow, more than 1 playbook to run on any incident. All execution logs should be retained and available for the reference.
68	Solution should allow differentiation between alerts and incidents (incident could be made of multiple alerts.)
69	The solution should visualise individual log source wise dashboards.
70	The solution must provide periodic updates of playbooks, OEM supplied Integrations and threat intelligence for incident artefacts.
71	The solution must support the ability to take-action related to an incident. For example, the solution should support the ability to block an intruder.
72	The solution must support the ability to correlate against 3rd party security data feeds (i.e. geographic mapping, known botnet channels, known hostile networks, etc.). These 3rd party data feeds should be updated automatically by the solution.
73	The system should be able to extract IOCs from PDF/csv/emails/other formats and search for those IOCs within the organization raw data. In case IOC is found, the system should trigger a new alert and save the indicator information in the local IOC Database.
74	The system should support creation of an incident based on an email input (e.g. analyse all emails from a dedicated phishing mailbox)

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75	The system should have an option to edit and change the event properties (for example its severity).
76	The solution must allow users to take remedial steps directly from within the visualization of incident correlation enabling a rapid and efficient response.
77	Solution must provide for a dashboard for virtual War Room/ collaboration platform on a per incident basis for comprehensive collection of all investigation actions, artefacts, and collaboration at one place
78	The solution must provide reporting templates, to report on incident information, for the management team as well as the IT Security team via the GUI. Describe how the solution provides the ability to configure reports.
79	The solution must provide configurable and customized report creation feature. Please describe how your solution meets this requirement.
80	Solution should provide for documentation of evidence like IOCs, messages, running analysis on artefacts, notes, adding artefacts, etc.) for later use for investigative purposes.
81	Solution should record timestamp for all actions taken in an incident
82	Solution should document all manual tasks perform by user in an incident
83	Solution should provide Predefined reports
84	Solution should support creation of customized reports in formats like csv, doc and pdf with custom logo of the organization
85	Solution should support Dashboards which can provide high level view of Platforms KPI's to the management
86	Solution should have documentation readily available for using automation and creation of custom automation
87	Solution should provide advanced Dashboards and reports based on KPI's to be tracked
89	The solution must deliver customizable dashboard widgets that can present relevant incident information to the users.
90	The solution must support a web-based GUI for management, analysis and reporting.
91	The solution must provide central management of incidents and administrative functions from a single web-based user interface.
92	Case Management: The proposed solution must provide complete process framework for integrating security monitoring and investigations with existing workflow procedures. Workflow should involve escalating an incident to other users within the same team or within other teams etc.
93	Workflow: The proposed solution should allow for assigning security analysts to specific security incident investigations. The proposed solution must provide a complete audit trail and accountability during the incident handling or forensic investigations. It should support the retrieval of relevant packets to a cyber-security incident.
94	Incident Tracking: The proposed solution must provide necessary tools to identify, isolate and remediate incidents as they occur.

95	The solution should provide a web based tool for incident management and the same should follow industry best practices
96	The administrator must be able to define role-based access to various functional areas of the solution. This includes being able to restrict a users access to specific functions of the solution that is not within the scope of a users role including, but not limited to, administration, reporting, incident assignment, playbook creation.
97	The SOAR solution should have inbuilt test lab.
98	SOAR should have an integrated Threat Intelligence Platform (TIP) and should Facilitate importing and parsing structured and unstructured intelligence documents- Structured/finished intelligence analysis reports (.txt, .PDF); Automatically ingest email lists with threat information; Formatted CSV Files, XML-based structured intelligence – STIX
99	TIP should De-duplicate indicator input data when imported from multiple sources; Provide features to add context to and enrich threat intelligence-Ability to rank or assign severity of risk to intelligence and IOCs
100	Solution must provide out of the box at-least 50 open source threat intel feeds and at-least 1 Premium feed
101	Support Integrations with Security Products-Native support for STIX/TAXII integrations, Export threat intel data with secure API, Integrate with additional tools and information sources via RESTful API
102	The solution should have out of the box capability to integrate with CERT-In CMTX threat feeds, NCIIPC TDP feeds etc.

Network Behaviour Anomaly Detection (NBAD)

Sr No	Specifications
1	NBAD should be capable of monitoring and managing minimum of 3 separate zones.
2	The solution should have single centralized dashboard
3	The tool should detect malware and botnets.
4	The solution should detect activities related to Advanced Persistent Threats (APT) and Trojans.
5	Ability to detect the presence of an Internet worm on the network based on propagation behaviour on a particular service port.
6	The tool should be able to perform Network Traffic Pattern Analysis based on IP addresses, groups of IP addresses, source/destination IP pairs and Bandwidth Analysis etc.
7	The tool should be able to perform Real time monitoring of host behaviours and traffic analysis to identify threats.
8	The tool should detect common events like DDoS / Do S, Scanning, Worms, Unexpected application services. (e.g., tunnelled protocols, backdoors, use of forbidden application Protocols), Policy violations, etc.
9	The tool should be able to identify from where the attack originated.

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10	Bidder shall suggest mitigation measures for security events like Denial of Service, DDoS etc. on real time basis.
11	a. Do mention the list of events that can be detected and action it can take, if any. b. Describe how tool can take care of slow and small attack over a prolonged period. c. Also mention overall limitations.
12	Should detect long-lived connections that may be associated with data-exfiltration.
13	Should detect applications running on non-standard port numbers.
14	The tool should be able to identify ports and services not necessary for normal business operations.
15	The events generated by the system should be classified at various risk level like High, Medium, Low etc.
16	The tool should have the capability to analyse physical and virtual network environments.
17	The tool should be able to do Centralized and non-intrusive traffic and flow-based network monitoring.
18	The solution should be able to Detect Anomalous Security Events and have threshold-based alerting.
19	The tool should have customized rule creation options
20	The solution should identify network traffic from potentially risky applications (e.g. file sharing, peer- to-peer) etc.
21	The solution should support traffic profiling associated with logical network design (e.g., Subnet/CIDR).
22	The solution should be compatible with industry leading appliances (like Firewall, IPS, VPN, routers, etc. for network flow analysis)
23	The solution must support and be scalable to cover assets of SLDC catering to distributed deployment.
24	The solution must be capable of identifying suspicious or hitherto undiscovered communication patterns. The solution must support detection of newly discovered pattern in future.
25	The solution should be integrable with other components viz. SIEM, VM, AD, LDAP, Firewalls, IPS, VPN, Routers, Switches, Internet Proxy, etc. List of supported technology and products and products/technologies not supported if any.
26	The system should be able to capture NetFlow and Packets.
27	The solution should identify applications using ports other than the well-known, and applications tunnelling themselves on other ports (e.g., HTTP as transport for MS-Instant Messenger should be detected as Instant messenger - not HTTP).
28	The solution should profile traffic by TCP and UDP Port
29	The solution shall provide application bandwidth utilization graph.
30	The solution should display traffic profiles in terms of packet rate.

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31	The solution should profile and present information in multiple timeframes. Profiles should be available for week, day and hour
32	The solution shall be able to display the new hosts coming into the Network. It should be able to take inputs from other systems (like, vulnerability scanners) and alert about the new hosts that are vulnerable.
33	The solution should display the type of data being transported via HTTP into and out of the network
34	Should be able to get regular feed from global threat intelligence for proactive monitoring and alerting.
35	All the Event, Alerts and other information pertaining to SLDC'S NG- SOC must remain within SLDC premises only. Any information moving out of SLDC premises shall be reviewed and approved by SLDC on need basis.
36	The solution must retrieve forensic results at lightning speed. Should retrieve atleast 5M records within 180 seconds
37	The solution should have machine learning capability for unlimited host without any additional licence
38	The solution must have the ability to capture network traffic.
39	Solution should allow investigation of logs and raw packets data from a single console
40	Proposed solution should help the analysis in various Network Threat Detection use cases as listed below, but not limited to a. Continuous Monitoring: Ability to capture network traffic, index and play back all network data, and to provide Analyst with timely, targeted and prioritized information b. Detect Suspicious traffic to an Unknown Country: Meta should provide all details in terms of Country name, Business context, IP - From and To Information. Analyst should be able to isolate such traffic swiftly and investigate the payload for any malicious egress or ingress. It should also have provision for analysts to integrate with Sandboxing environment for malware analysis.
41	Solution should store RAW packet DATA for 7 days and normalized packet data for 15 days for forensics.
42	Should Capture at-least 512 Mbps of internet traffic at DC.

User and Entity Behaviour Analytics (UEBA)

Sl. No	Specifications
1	Support for creation of Watch Lists for suspicious users
2	On-premise deployment
3	Said UEBA tool should be able to integrate with proposed SOAR & SIEM solution and should be from same OEM for monitoring user endpoints and reporting anomalies

4	The solution should be able to highlight risky and potentially abnormal user
5	The solution must be able to adjust itself in DHCP environment and yet uniquely identify machines with high accuracy despite change of IP Address
7	The solution should have permission for device admin, subnet admin.
9	Should be able to show us RAW and Normalized logs, or relevant data basis on which anomalous behaviour was observed
10	The solution should allow administrators to setup of device label, device priority, alert by priority, alert by score
11	Integration with enterprise authentication/SSO platforms for simplified access
12	Availability of out-of-the-box administrative dashboards and reports
13	Ability to monitor, report and alert on the data streaming in and out of the system
14	Identity based threat plane behaviour analysis for account hijacking and abuse
15	Proactive and actionable alerting for anomalous behaviour and risk scores
16	High privilege access anomaly detection for misuse, sharing, or takeover
17	Tracking of user in case of IP switch
18	Identifying Trusted Host and Entity Compromise instances
19	Uses self-learning behavioural analysis to dynamically model each device, probabilistically identifying any anomalous activity that falls outside of the device's normal pattern of life.
20	Unusual Credential use - models the times and devices normally used by each username, and alerts when there is an unusual combination
21	Use of supervised machine learning algorithms
22	Flexibility to configure rolling window of period for behaviour profiling
23	Dashboards for different roles and access levels
24	Customizable dashboards, configurable policies and risk model optimization
25	Provide various visualization options for deep-dive investigation, compliance and reporting
26	Ability to perform detailed search on raw and enriched data
27	Support natural language search capability without requiring to learn a custom search query language
28	Intelligent search providing auto-complete, auto-suggest capabilities based on contextual data
29	The solution should allow network data ingestion from log collectors for Netflows to provide Network Anomaly Detections
30	The solution should be installed passively into infrastructure

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31	The solution should be able to automatically identify and classify users and entities (i.e. devices, applications, servers, data, or anything with an IP address)
32	Support highly available component architecture ensuring no single point of failure
34	Exporting and report generation capabilities (Excel, PDF, XML)
35	Availability of out of the box reports for analysts to make investigation decisions
36	Availability of out of the box reports for administration and management
37	Availability of out of the box reports for audit and compliance
38	Ability to create custom reports and schedule the same
39	Reports can be delivered as CSV, Email, PDF
40	Ability to schedule reports with periodic intervals
41	The solution should provide coverage into virtual environment
42	Access high-value assets: User starts accessing and downloading high- value assets with increased frequency.
43	Usage changes over time: User activity deviates from normal over a short period of time or a gradual change over an extended period of time
44	Usage deviates from peer group: User pattern of activity starts deviating from the peer group
45	Change in account privileges: User attempts to change privileges on existing account or open new accounts on other systems
46	Application misuse by sequence of actions: User performs a sequence of actions which no other user is performing
47	Sensitive data leakage: User manipulates http request / response parameter to download sensitive data
48	UEBA should activate rules for a set of users until a specified condition or specified time window
49	More data being transferred then a normal to and from servers and/ or external locations
50	Should identify Account used for the first time in a long time
51	Should identify User involved in previously malicious or threatening behaviour
52	Detect insider threats, account hijacking and abuse, plus data exfiltration
53	Work-centric UI with case management, or input to third-party solutions
54	The solution should consist of a powerful visualization platform that enables threats being analysed and investigated intuitively
55	The solution should be able to administer from a web browser
56	The solution's UI should be able to provide a real-time, operational overview of an organization's entire network and the threat level it faces at any given time

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57	The solution's UI should allow displaying threat by user, device and model with sorting and selected period
58	The solution's UI should be able to display particular events in a graphical way
59	The solution's UI should allow checking the device summary and similar device behaviour
60	The solution's UI should allow using advanced search for detail connection information. Certain application-layer protocols are further decoded during this process
61	The solution's UI should provide a Google-like search bar to search a device by Hostname, Mac Address, Username of user logged into that device, IP Address, Nickname.
62	Generate a threat Report which will look over a specified time period and produce a PDF based on the statistics generated.
63	Allows us to create Incidents out of Events/alerts onto which analysts will collaborate inputs and for which, reports can be exported
64	Multiple elements can be correlated into an incident, from metric graphs to device log entries.
65	System and application logs show who did what, at what time, within the UEBA application
66	The UEBA solution should be from the same SIEM and SOAR OEM.
67	The entire Event, Alerts and other information pertaining to Organization's SOC must remain within Organization premises only. Any information moving out of Organization premises shall be reviewed and approved by Organization on need basis.
68	Machine learning should support following types of data science models at min.: 1) Time based models 2) Discrete Models 3) Categorical Model 4) Continuous Model
69	Platform should be able to create profiles and monitor all Users in AD from day 1.
70	Solution should allow users to consider any field as entity and map it for anomaly detection.

DAM (Database Activity Monitoring)

S. No.	Functional Specification
1	Unified DAM Platform: The solution should be a single console integrating DAM, Vulnerability Assessment, and Data Classification with one identity and alerting pipeline.
2	Flexible Deployment: Should support on-premises and air-gapped deployment from a single codebase.
3	Broad Database Support: Should support multiple OEM database as well as provision for integration with any custom database as well, like Oracle, SQL Server, MySQL, PostgreSQL, Db2, MongoDB, etc.
4	The solution should be available for major OS and should use native database audit interfaces without kernel modules, packet interception, or database restarts.
6	Solution should capture DDL, DML, DCL, TCL, authentication events, administrative changes, stored procedures, RPC/database links, and privileged-user activity.

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7	Solution should also filter by user, role, schema, table, statement type, and time window, with configurable rate throttling and pause/resume capability.
8	Solution should records DB user, OS user, application user, client IP/hostname, session IDs, nanosecond timestamps, full SQL text, parameters, object metadata, execution result, timing, and row counts.
9	Solution should append-only event store with tiered retention.
10	solution should have tamper-evident audit trail with HMAC-signed events, hash-chained batches, immutable meta-audit logs, and prohibition of event deletion by any role.
11	Solution should have real-time policy evaluation like Sub-second evaluation with built-in policies for privileged misuse, brute force, bulk extraction, SQL injection, and change-window violations.
12	Solution should support advanced rule management
13	Solution should support free-text and structured search, timeline visualization, user/object/session pivots, case management, signed PDF evidence exports, and chain-of-custody bundles.
14	Solution should have built-in integrated database VA
15	Solution should have automated sensitive data discovery schema.
16	Should have RBAC for different roles.
17	Solution should have Pre-Built Compliance Reporting
18	Solution should have Real-time alerts via Email, SMS, Slack, Microsoft Teams, webhooks etc. with escalation, suppression, grouping, and acknowledgement workflows.
19	Solution should integrate with the ITSM tool and should be from the same OEM as NMS & ITSM.
20	Solution should be able to integrate with the SIEM
21	Solution should support enterprise authentication like local auth, LDAP/AD, etc.
22	Solution should have cryptographic security
23	Solution should work in HA.
24	Solution should have backup provision and data to be kept for atleast 90 days
25	Solution should be web-console based to enable automation, integration, and accessible administration.
26	Solution should have multi-tenancy
27	Solution OEM should comply with ISO 27001, ISO 27034, ISO 9001, ISO 14001, CIS etc.

2x2 Videowall/Display

Sl No	Technical Specs	Minimum Specifications
	General	
1	Size (diagonal)	115 Inches
2	Technology	LED
3	Aspect Ratio	16:09
4	Resolution	3840X2160
6	Maintenance Mode	Yes
7	Installation Mode	Wall-Mount

	Optical Parameter	
8	Brightness (Typ.)	1000 cd/㎡ (Adjustable)
9	Refresh Rate	120Hz
10	Gray Scale	16 bit
11	Contrast	5,500:1
12	Color Temperature	
13	Viewing Angle	Horizontal 178° or higher , Vertical 178°
14	LED Lifespan	50,000 hrs
	Electrical Parameter	
15	Operation Voltage	100~240VAC 50/60Hz
16	Standby Power	0.5W
17	Average Power Consumption of Overall Unit	359(W/H)
18	Maximum Power Consumption of Overall Unit **System Parameter**	836 (W/H)
19	Operating System	Tizen
20	CPU	Quad code
21	RAM	4 GB
22	Storage	16 GB
23	Media Format	Support mainstream video format such as MPEG-1,MPEG-2,MPEG-4,H.263,H.264,H.265,VC-1,VP9,VP8,MVC,AV1; Support audio formats such as MP3; Support JPG, JPEG, BMP, PNG, GIF and other image format.
	Environmental Parameter	
24	Operation Temperature	-10℃~40℃
	I/O Ports	
25	HDMI IN	3
26	USB 2.0 IN	2
27	Audio Output (3.5mm Jack)	1
28	Control	YES
29	Wifi	YES
30	Bluetooth	YES
31	Ethernet	YES
	Speakers	60W 4.2.2CH
	OPS	
32	CPU	Intel® Core™ I5-1240P
33	Memory	Dual channel DDR4,16GB(8GB*2) Integrated graphics
34	SSD	1×M.2-2280 SATAIII SSD,256GB
35	Ethernet	1×RJ45(10/100/1000M)

36	System Version	Windows
	I/O INTERFACE	
37	HDMI IN	2
38	HDMI OUT	1
39	USB 3.0	2
40	1000M RJ45	1
41	PC Interface	40pin
	Accessories	
42	Remote Control	1
43	Front Maintenance Tool	1
44	Installation Tool kit	1
45	Wall-mounting Bar	1
46	Ceiling mounted	optional
47	Mobile Floor Stand	optional
	TOUCH FEATURE	
48	Touch type	Laser Radar
49	Sampling frequency	50 kHz
50	Scanning frequency	20 Hz
51	Touch point	Single
52	The height of the raised border	≤ 1mm
53	Fit size	120", 138", 150", 165"
54	Minimum object size	6mm
55	Accuracy	≤ 10 mm
56	Touch Height	≤ 20 mm
57	Response time	≤ 300 ms
58	Warranty and Support	5 years OEM warranty - 5-year parts, 5-yearlabor, 5-year onsite support.

24*7 Workstation

S.No.	Particulars	Specifications
1	**Processor**	14th or latest generation Intel Core i5 or Higher. Offered Processor must be from desktop segment only (mobile segment processor will not accepted)
2	**Chipset & Motherboard**	Intel Q670 Chipset or higher
3	**Memory**	16 GB DDR5, 4800 MHz RAM or higher, expandable up to 64 GB or better
4	**Graphics**	Integrated Intel Graphics or better
5	**Hard Disk Drive**	M.2 512GB PCIe NVMe SSD
6	**Display**	23.8" FHD, Anti-Glare, IPS, TUV Low Blue Light certified
7	**Security**	Discrete TPM 2.0 with Chassis Intrusion Switch

8	**Keyboard**	Wired Keyboard (same make as PC)
9	**Mouse**	Wired Mouse (same make as PC)
10	**Networking facility**	Integrated Ethernet LAN 10/100/1000 and Wi-Fi 6 + Bluetooth
11	**Ports & Slots**	1 RJ-45 port 10/100/1000, 6 USB 3.2 ports and 1 USB Type-C port, 1 Universal audio jack, 1 Line-out audio port, 1 DP Port, 1HDMI In & 1 HDMI Out Ports, 3 M.2 slot, 1 security-cable slot
12	**Audio**	Integrated dual Internal speakers and Mic
13	**Webcam**	1080p FHD Webcam
14	**Form Factor**	All In One Form Factor with Stand
15	**Operating System**	Microsoft Windows 11 Professional 64 bit OS or better
16	**Performance Enhancer Software**	AI-based performance optimization software to optimise and enhance the performance of Audio, Applications and Network
17	**Certification**	Ubuntu Linux OS Certification, Windows 11 Certification, EPEAT® India registered, MIL STD 810 G/H Test Passed, Energy Star,
18		For OEM : ISO 9001, 27001, 14001
19	**Power Supply**	Maximum 200W Integrated Power Supply
20	**Accessories**	To be supplied with required accessories, connecting cables and driver media & power cable.
21	**Warranty**	5 years on site comprehensive Warranty

Necessary Server & Storage infrastructure

Components	**Server Specifications**
Form Factor	Maximum 1U Rack mount server
CPU	Server should have latest generation 64 core processor supporting hyperthreading with minimum 3GHz base frequency or better
RAM	Server should have 512GB DDR5 RAM populated at day 1 and should be scalable upto 1.5TB at least without replacing existing DIMMs
Network ports	4 x 25Gig fiber ports populated with SFP28 SR optics supporting port level redundancy
FC ports	2 x 32Gbps fiber ports populated with SR optics supporting port level redundancy
PCIe slots	Server should have at least 2 x PCIe Gen5 slots available for future use after populating required ethernet and FC ports asked
Internal Storage	Server should have minimum 8 SSD drives bays populated with 2 x 480GB SSD drives in RAID 1 configuration at day 1
RAID Controller	Integrated/PCI based 24Gbps HW RAID controller with minimum 8GB cache supporting RAID 0,1,10,5,6
Power Supply	Offered server must have ≤ 1500w N+N redundant power Supply of atleast Platinum grade
System Security	Server should have below security essentials and certificates:

Recommended

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	1. shall include FIPS 140-2, TPM 2.0 v5 & UEFI with secure boot.
	2. real time firmware security scanning to protect against firmware which executes before the OS boots.
	3. maintain repository for secure firmware lifecycle management with rollback capability, if required.
	4. Firmware and software updates/upgrades shall support cryptographic signature verification to ensure installation of only authenticated and trusted releases from the OEM..
	5. End-to-end supply chain controls to ensure component integrity, security, and vendor responsibility.
Security Bezel	Security bezel with locking function
Intrusion alert	Intrusion alert in case chassis cover being opened
Management integration	Support for integration with Microsoft System Center, VMware vCenter, BMC Software
Power & temperature	The server management solution shall provide real-time power monitoring, power consumption history, threshold-based alerts, and graphical visualization of power and thermal parameters.
	Temperature monitoring & graphing
Pre-failure alert	The server management solution shall provide hardware health monitoring and proactive alerts for actual or potential failures of critical components including fan, power supply, memory, CPU, RAID controller, NIC and storage devices.
Management port	Dedicated 1GbE Base-T management port through onboard BMC. The server management solution shall be provided by the OEM and shall support required management, monitoring, security and remote administration features.
Server management	Embedded management controller supporting real-time monitoring and telemetry through standard APIs such as Redfish/IPMI
	Management platform shall support telemetry export and integration with external monitoring, observability, and SIEM tools using open standard APIs/protocols (Redfish/REST API/SNMP/Syslog or equivalent.
	Integration with third-party monitoring and AIOps platforms through standard APIs.
	Rack-level management must include native support for an integrated rack controller, with policies that automatically initiate coordinated server shutdown on critical rack events such as liquid-leak or power faults.
	Management platform shall support integrated power and thermal monitoring with real-time telemetry, alerts, and energy utilization reporting through standard APIs.
	The management solution shall support centralized firmware inventory, update and lifecycle management for server components including BIOS, BMC, storage controllers, network adapters and other supported hardware components.

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	The platform must provide officially supported automation integrations for both Ansible and Terraform, enabling full provisioning, configuration and update workflows to be driven via code.
	Advanced automation and integration features must be enabled via a per-device, perpetual license option, allowing customers to selectively activate these capabilities only on chosen servers while managing licenses centrally from the same console.
Security Requirements for Management Platform	The embedded management controller shall support hardware-based secure boot and digitally signed firmware validation to ensure execution of only authenticated OEM firmware.
	The management stack must enforce strong, centrally managed authentication policies, including directory integration (for example, LDAP/AD), multifactor authentication options and granular, role-based access control down to specific tasks and device groups.
	All management traffic (web, APIs, CLI, telemetry, backup/restore and log forwarding) must use modern TLS with configurable cipher suites, and the console must reject SHA-1 and similarly deprecated certificate algorithms.
	The management solution shall support monitoring of server and controller configuration, detection of security configuration deviations, and policy-based remediation of non-compliant settings.
	The management solution shall support export of alerts, audit logs and configuration change records to external Syslog/SIEM platforms using secure TLS communication.
	The management controller shall support secure backup and restore of configuration data using encrypted protocols such as HTTPS/SFTP. Access to backup and restore functions shall be controlled through role-based user privileges.
Rack kits	Rack mounting with sliding rails
Certification	OEM need to submit latest and valid FCC , CE Certification & BIS Certificate
Server Hardware Warranty	5 Years comprehensive NBD On-site warranty with 24x7x365 remote hardware support. warranty should reflect in the support web site of the OEM
Installation	Installation and knowledge transfer on supplied infrastructure
Drive media retention	Faulty drives will not be returned during the warranty period

Firewall in HA

S.No	General Requirement for Firewall	
1	Type	NGFW (Next Generation Firewall)
2	Form Factor (RU)	1 RU
3	Features	Layer 3-Layer 4, NAT, VPN, Application Visibility and Control (AVC), User Identity, Next Generation Intrusion Prevention System (IPS), Zero Day Protection/Advance Malware Protection, Web Security Essentials/URL Filtering
4	Traffic handled	TCP, UDP, HTTP/TCP, TCP/UDP

7	NGFW Throughput (Real World/ Prod Performance) (Gbps)*	9 Gbps of higher
8	IPSEC VPN Throughput	5Gbps
9	Concurrent Session/ Concurrent Connection	1.2M or higher
10	New session/Connection per second (Layer 7/app-id connections per second)	80K or higher
11	Type of Interface Supported	GE Copper/10G SFP+
12	Number of GE Copper interface	8 or higher
13	Number of 1/10G SFP+ interface	8 or higher with additional slot for future expansion
14	Number of VPN Peers supported (Site to Site)	-min 1500
17	Type of Storage Disk	SSD
18	Storage Capacity (GB)	min 900 GB and Spare slot
19	Memory	min 32GB
20	Power Supplies	Dual
21	Hot Swappable Power Supply	Yes
22	Hot Swappable Redundant Fan	Yes
23	Type of Processor	x86
24	High Availability Support	Yes
25	If Yes, High Availability from day 1	Any Value
26	Firewall Policies - License	Yes
27	Details of the Firewall Policies for the Firewall provided with the License	Web Security Essentials/URL. Filtering, IPS License, Application Visibility License, APT(Advance Persistent Threat) License (Anti Malware Protection, C&C attacks, Geo IP Protection, Zero Day Threat Protection), Gateway Antivirus, Gateway Antispam
28	Security Intelligence	IP, URL, Domain, Passively detect End Point, IOC Intelligence
29	IPv6 Ready from day 1	Yes
30	OEM Warranty(Year)*	5 or higher
31	The firewall should supports NAT's like source NAT , destination NAT , U-Turn NAT. Firewall should support Nat66, Nat 64 & Nat46 functionality.	Yes
32	Solution should provides capabilities like dynamic real-time metrics based , policy-based, application path selection across multiple WAN connections and supports service chaining for additional services such as WAN optimization and firewalls.	Yes
33	The solution should provide the following routing capabilities: OSPF, BGP, RIP, Multicast, Static, Route Tracking(SLA) PBR, ISIS, BFD, ECMP, VRF, Application based Routing	Yes

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34	The solution should be able to provide contextual information about the hosts and the network subnets present such that the admins are able to capture all the required information and build the security profiles based on the details shown on the solution . The details captured should consist of the following: IOC's , MAC addresses, IP address , Applications , Ports and protocols, vulnerabilities etc.	Yes
35	Should support capability to integrate with other security solutions to receive contextual information like security group tags/names.	Yes
36	Should support more than 5000+ (excluding custom application signatures) distinct application signature as application detection mechanism to optimize security effectiveness and should be able to create 40 or more application categories for operational efficiency	Yes
37	Should be capable of dynamically tuning IDS/IPS sensors (e.g., selecting rules, configuring policies, updating policies, etc.) with minimal human intervention.	Yes
38	Should support more than 25,000 (excluding custom signatures) IPS signatures or more. Solution should be able to passively detect endpoints and infrastructure for threat correlation and Indicators of Compromise (IoC) intelligence. The signatures should also have categorization based on MITRE TTP's.	Yes
39	The firewall solution should have the following capabilities to make sure the current solution is future ready for technologies like WAN routing , SASE etc. The firewall should have application aware routing with HTTP and ICMP ping , ZTNA based clientless access to applications.	Yes
40	The firewall solution should have capabilities like Application Aware Routing, Health Monitoring, DIA, Dual ISP, Data interface Management for simplified branch capabilities	Yes
41	Should be capable of automatically providing the appropriate inspections and protections for traffic sent over non-standard communications ports.	Yes
42	The solution should be able to identify, decrypt and evaluate both inbound and outbound SSL traffic on-box. The NGFW shall support the ability to have a SSL inspection policy differentiate between personal SSL connections i.e. banking, shopping, health and non-personal traffic.	Yes
43	The solution should have ML/AI capability to detect client apps and process . Moreover it should be able identify malicious encrypted traffic even when it is destined for a trustworthy service. This is required to help administrators control specific applications and	Yes

	improve network security	
44	The solution must provide IP reputation feed that comprised of several regularly updated collections of poor reparation of IP addresses determined by the proposed security vendor. Solution must support IP reputation intelligence feeds from third party and custom lists of IP addresses including a global blacklist. The solution should have the capability to detect MD5, SHA256 and SHA512 traffic hashs to detect any malicious traffic pattern	Yes
45	The solution should provide Configuration Deployment History, Pending Changes and Policy Compare capability before the security policies are deployed on the firewall. It should also provide configuration rollback capability to the last good configuration running on the firewall.	Yes
46	Should must support DNS threat intelligence feeds to protect against threats	Yes
47	The Appliance OEM must have its own threat intelligence analysis center and should use the global footprint of security deployments for more comprehensive network protection.	Yes
48	The detection engine should support capability of detecting and preventing a wide variety of threats (e.g., network probes/reconnaissance, VoIP attacks, buffer overflows, P2P attacks, etc.).	Yes
49	Should be able to identify attacks based on Geo-location and define policy to block on the basis of Geo-location	Yes
50	The detection engine must incorporate multiple approaches for detecting threats, including at a minimum exploit-based signatures, vulnerability-based rules, protocol anomaly detection, and behavioural anomaly detection techniques.	Yes
51	Should support Open based Application ID / Custom Application ID for access to community resources and ability to easily customize security to address new and specific threats and applications quickly	Yes
52	Should must support URL threat intelligence feeds to protect against threats	Yes
53	Should support Reputation- and category-based URL filtering offering comprehensive alerting and control over suspect web traffic and enforces policies on more than 250 million of URLs in more than 75+ categories.	Yes

54	Should support the capability of providing network-based detection of malware by checking the disposition of unknown files using SHA-256 file-hash or signature as they transit the network and capability to do dynamic analysis.	Yes
55	Solution shall have capability to analyze and block TCP/UDP protocol to identify attacks and malware communications. At minimum, the following protocols are supported for real-time inspection, blocking and control of download files: HTTP, SMTP, POP3, IMAP, NetBIOS-SSN and FTP	Yes
56	Proposed solution shall have required subscription like Threat Intelligence for proper functioning	Yes
57	The solution should support IPSec and SSL VPN	Yes
58	The Solution shall provide a secure Remote Access VPN capability for users connecting from trusted and untrusted networks, using industry-standard SSL/TLS and/or IPsec protocols.	Yes
59	The proposed solution must support endpoint posture assessment (HostScan) at the time of VPN connection and during the active VPN session. The HostScan functionality shall be tightly integrated with the VPN gateway and shall not require third-party posture engines for baseline checks.	Yes
60	HostScan must be capable of checking, at a minimum: Operating system type and version Antivirus / Anti-malware presence, status, and update level Endpoint firewall status Disk encryption status (where supported) Presence of prohibited or mandatory processes/applications	Yes
61	The solution should provide 100 SSL VPN license from day1	Yes
62	The management platform must be accessible via a web-based interface and ideally with no need for additional client software and must provide centralized logging and reporting functionality	Yes
63	The management platform must be a dedicated OEM appliance and VM running on server will not be accepted	Yes
64	The management platform must be capable of integrating third party vulnerability information into threat policy adjustment routines and automated tuning workflows	Yes
65	The management platform must be capable of role-based administration, enabling different sets of views and configuration capabilities for different administrators subsequent to their authentication.	Yes
66	Should support troubleshooting techniques like Packet tracer and capture	Yes

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67	The management platform must provide multiple report output types or formats, such as PDF, HTML, and CSV. The management platform support running on-demand and scheduled reports	Yes
68	The management platform must support multiple mechanisms for issuing alerts (e.g., SNMP, e-mail, SYSLOG).	Yes
69	The centralized management platform must not have any limit in terms of handling logs per day	Yes
70	The management platform must provide built-in robust reporting capabilities, including a selection of pre-defined reports and the ability for complete customization and generation of new reports.	Yes
71	The management platform must risk reports like advanced malware, attacks and network	Yes
72	The management platform must include an integration mechanism, preferably in the form of open APIs and/or standard interfaces, to enable events and log data to be shared with external network and security management applications, such as Security Information and Event Managers (SIEMs), and log management tools.	Yes
73	Proposed solution should support 24x7x365 OEM TAC support and advance Next Business Day Hardware replacement	Yes

48 U Server Rack with accessories

1	Racks manufactured out of steel sheet punched, formed, Powder coated
2	Standard for Racks configuration will be frame, side panels and vented top cover
3	Rack should have Front Glass Door and Rear Steel door (Vented).
4	Rack should have 2 nos of removable side panel with slam latch.
5	Rack should have provision to mount racks on Floor
6	Rack should be 48U (1U = 44.45 mm) in Height.
7	It should be 800MM Wide,1000MM Deep and Overall height 2200 MM.
8	Rack should Conforms to DIN 41494 or Equivalent EIA /ISO / EN Standard
9	Rack should have Adjustable mounting depth,
10	Rack 4 No Adjustable, 19" verticals with Punched 9mm Square Hole and Universal 12.7mm-15.875mm-15.875mm alternating hole pattern offers greater mounting flexibility, maximizes usable mounting space.
11	Rack should have 100% assured compatibility with all equipment conforming to DIN 41494 (General industrial standard for equipment)
12	Powder coated finish with seven Tanks pretreatment process meeting IS
13	Rack should have Proper Grounding & Bonding
14	Rack should have Fan module Mount Provision on top Cover

15	Rack should have Fan tray with 4 no's 90 CFM Fan
16	Rack should have two Fixed shelf with 700mm depth for mounting NON-Rack mountable equipment
17	Rack should have two Horizontal Cable manager - PVC Finger Duct w/ Removable Cover,19", 1U, 60mm,
18	Rack should have two Server /IT Rack mount power distribution unit, 1Ph, 230V, 16A, 50/60Hz, 1U standard with 12 Socket x IEC C13, Inlet Plug type 16A Indian Round Pin, 16A MCB - PDU Rating 3.7KVA/Side feed-2.5Mt/ Black
19	Rack should have provision for cable entry Exit from both top & bottom.
20	Rack should have 1 Packet of mounting hardware, Pack of 20.
21	Warranty:- 05 years or higher

Switches/Router

S. No.	Specification
1	Shall be 1RU, 19" Rack Mountable fully managed switch
2	24x 10/25G Fiber Ports ports, 2x 40/100G fiber uplink ports.
3	1 RJ-45 (serial RS-232C) or USB console port
4	16 GB MB flash, 16 GB DRAM
5	Packet Buffer 16MB
6	Shall have switching capacity of 2 Tbps for providing non-blocking performance on all Gigabit ports and min 1 Bpps of forwarding Capacity.
7	Should support stacking or equivalent via dedicated stacking ports. Should support for minimum 200 Gbps of stacking thoughput with minimum 8 switch in stack. All the required cables/accessories for stacking should be provisioned from Day 1.
8	IEEE 802.1D STP, IEEE 802.1w RSTP and IEEE 802.1s MSTP, 802.1x, 802.1AE, 802.3x, 802.1p, 802.1Q
9	IEEE 802.3ad Link Aggregation Control Protocol (LACP) up to eight links (ports) per group
10	MAC address table size of 32K entries and 1K Active VLANs.
11	Shall support 7K ACLs, 7K Multicast Routes and 15K IPv4/IPv6 Routes
12	Shall support Jumbo frames to improve the performance of large data transfers
13	Internet Group Management Protocol (IGMP), IEEE 802.1AB Link Layer Discovery Protocol (LLDP), Multicast Listener Discovery (MLD) snooping and PIM.
14	BGP, IS-IS, VRF, VXLAN, OSPF, Policy-Based Routing and VRRP from Day 1
15	IPv6 host and Dual stack (IPv4/1Pv6) support to provide transition mechanism from IPv4 to IPv6
16	Traffic prioritization based on IP address, IP Type of Service (ToS), Layer 3 protocol, TCP/UDP port number, source port, and DiffServ. Should support strict priority queuing.

17	Access Control Lists (ACL) for traffic filtering and Dynamic VLAN assignment
18	Should support MACSEC as per IEEE 802.1AE on hardware from day 1.
19	Should support port security, DHCP snooping, Spanning tree root guard, First Hop Security or equivalent.
20	Switch should capable to understand that system OS are authentic and unmodified, it should have cryptographically signed images to provide assurance that the firmware & BIOS are authentic.
21	Configuration through the CLI, console, Telnet/SSH and browser-based management GUI.
22	SNMPv1, v2, and v3 and Remote monitoring, (RMON) support, sFlow/netFlow or equivalent for traffic analysis
23	TFTP and Secure FTP support.
24	RADIUS/TACACS+ for switch security access administration
25	Network Time Protocol (NTP) or Precision Time Protocol (PTP)
26	Should support comprehensive programability features like Model-Driven Programmability, NETCONF/REST interface, API or equivalent.
27	2 field-replaceable and hot-swappable power supplies from day 1
28	Operating temperature of 0°C to 45°C
29	Switch shall conform to UL/IEC/EN 60950-1 or latest UL/IEC/EN 62368-1 Standards
30	Switch / Switch's Operating System should be tested for EAL2/NDPP/NDcPP or above under Common Criteria Certification.
31	Warranty:- 05 years or higher

Network Management System

Sl. No	Specifications
1	System should have option for multiple options for discovery including IP address based discovery, IP address range discovery, CSV based discovery for bulk discovery and it should allow options to add custom fields to support customer specific data to upload during discovery
2	The system should fetch topology via SNMP for ARP tables from routers , MAC tables from layer 2 switches, cisco Discovery Protocol, Link Layer Discovery Protocol, Foundry Discovery Protocol or SynOptics Network Management Protocol. The discovery should be automated and continuous.
3	Discovery has to work intelligently by identifying the device in the network by the given IP range and categorize into network devices and servers with vendor and model details.
4	Automatically learn devices that supports SNMP, HTTP, Ping, SMTP, POP3, WMI,JMX, SOAP, REST API,PDC, SSH and Telnet along with any required protocol to communicate to the devices.
5	System should support global threshold and it should have option to define individual resource/interface statistics level threshold

6	Threshold & Alerts: define and store static & dynamic resource utilisation thresholds, define and maintain availability thresholds for individual components, services or applications, individual service actions / automated operations (performing a transaction, running an automated procedure etc.), support comparison of actual availability , actual performance with agreed levels and alerting on breaches
7	Configurable parameters like frequency, data duration, resolution duration, sigma based polarity value, reset points should be available
8	All thresholds should have set point , reset point, polarity , set point message and reset point message for ease of use.
9	Detect & highlight faults (abnormal situations) in near real-time occurring anywhere within the monitored IT Infrastructure
10	Provides Filtering, De-duplication, Holding, Suppression and Correlation capability to let user focus on the critical event that affects the business and business processes
11	Provides multi-level (preferably six-level) Severity definition, will handle events automatically and inform the designated person as per operational requirement
12	System should support separate Rule Engine based alarms apart from the generic threshold. a. Should have capability to configure Device Group based, Node Based, Resources/Interface based, Aggregation link based. b. On Selection of Nodes/Resources/Aggregation links it have flexibility to filter based on fields available in node information c. Rules should have option to apply configuration on top of performance value or based on configured threshold alarms d. Rules should have option configure the breach based on min, max and average values e. Should have option to configure rules n repeat counters f. Should have options to select custom alarm and clear alarm messages for individual configured rules g. Should have option to send severity levels like error, warning and information h. Notifications support based on configured rules
13	Provides alarm suppression with hold time and aid in prevention of flooding
14	Sends alert via E-mail, SMS, Execute Batch file, SNMP Trap, XML notification, Pop-up window and Audio alert
15	Capacity Reservations: tool should allow management of resource allocations and reservations (for services, applications or other needs), identify resource shortages and provide information for further analysis or procurement
16	Event Record & Classification: possible to generate event for all the monitoring devices, tool be used to define thresholds to generate events, collect from 3rd party using REST API , on regular interval Polling API and collect events from 3rd party system, classify them , assign different levels of severity to events, attribute events to specialised practices such as performance and capacity management, availability management, information security management, allow the automatic recording of event messages reported by external tools, filter & differentiate between

	different types of events
17	Configurations: create rules that automatically assign deadlines to events based on their impact on services or on end-users, create rules that perform automated assignment of events to the corresponding teams, create rules that control automated notification of interested stakeholders about events , automatically handling duplicate events, provide event correlation capabilities to combine a set of different events into one major event
18	Monitors all traffic from all the interfaces of the network device. Provides traffic Utilization based on individual interface level, nodes level or based on the group by location, branch, departments etc.... as an Avg, Min and Max bandwidth, utilization, throughput or any custom monitoring parameters.
19	Provision to change the polling interval to any frequency depending on the priority till the individual component / resource level like each interface might have the different polling interval in the same device based of the criticality and importance of service customer
20	System should have capability to configure business , non-business hours or custom time polling. These configuration should be available for every device as well as every component in the device.
21	Provision to disable and enable the polling of specific type of devices
22	System should have capability to configure the maintenance period for any device. When device is in maintenance period there is no polling done and the SLA clock on the device is stopped.
23	SLA calculation / Isolation report should be made with the consideration of both the Primary and Secondary link together instead of individual link based. The downtime calculation will be measured when both the links are down for internal reporting and link based for ISP reporting. System should provide the flexible configuration in UI itself based on user needs
24	Provide a notification mechanism that allows administrator to define what notification channel to be used in different time of days, and able to trigger multiple notifications to alert multiple person and actions
25	Provide standard reports that display current status of nodes and interfaces. Reports could be viewed on daily graph (5 minute average), weekly graph (1 hour average minute average), monthly graph (1 hour average) and yearly graph (1 day average)
26	Reports & Dashboard: Pre defined Availability, Performance , Capacity Dashboards & Report, Should have option to drag and drop dashboard configuration option Should have option to add multiple type of widgets along with (Tabular, Summary, Multiple Graph options), Filter (predefined & Dynamic), Sort, Search, Group by, multi threshold, Export option to PDF , XLS etc., Generate automatically and send over Email or to a predefined folder, possible to define custom KPIs , A KPI definition must include the period-bound target value and the threshold value (to distinguish between the Green, Amber and Red zones).
27	Automatically generate daily reports that provide a summary of the IT Infrastructure as well as custom Reports and that are automatically sent by email at a pre-defined schedule to any recipient or save into any specific folder or drive.

28	Supports instant diagnosis of the node status through Ping, Telnet and SNMPwalk
29	Support Real-Time report generation for checking continuous reachability of target device
30	System should provide many different types of topology representation. To perform the following: 1. Display physical connections of the different devices being monitored in the system 2. Display flat maps of the entire network or networks in a single view 3. Display customer maps based on user configurations 4. Display maps based on geo locations
31	Automatically learn IP Networks and their segments, LANs, hosts, switches, routers, firewalls etc. and to establish the connections and to correlate
32	Provides provision to draw & map user specific network diagram
33	The tool should have Integrated Web based feature to build Network Diagram, No separate client window to configure network Diagram. The builder should be similar to MS Visio with all pre-loaded shapes and icons.
34	It should be a Drag & Drop based Network Diagram builder, Dynamically Upload Images, Customizable objects to support multiple vendors, capability to export maps in an XML format and upload to any other system.
35	Panel View a. Panel view should look similar to the actual device front panel b. System should automatically detect the device model display the right panel without any additional configuration c. Panel should show all the monitored interface with status d. Fan status with live fan icon and LED status for power
36	Tool should have complete inventory information of the assets discovered along with an option to fetch the target network device EoL / EoS information if required
37	Tool should allow QoS monitoring of WAN links across multiple technologies like Cisco IPSLA, Juniper RPM, Huawei NQA etc. across multiple protocols like HTTP, TCP, FTP, DNS etc.
38	QoS parameters should include link response time, link-level latency, link-level packet loss, link-level jitter, Round-Trip-Time etc.
39	Should monitor Class-Based Quality of Service (CBQoS) to find out if traffic prioritization policies are effective and if business-critical applications have network traffic priority. Should also support CBQoS Nested policies
40	Tool should have option to collect and store system logs from target devices including firewalls, routers, switches, WLC, servers, applications & databases
41	Tool should have multiple filtering options for incoming system logs based on target device, log_ID, severity, level, message, OS type, application / database etc.

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42	Tool should have option to export specific syslog messages to users via email / SMS
43	System should support VM, Hypervisor and Cluster monitoring from different vendors like VMWare, Citrix, Nutanix, Linux etc.
44	System licensing should be based only on Physical Hosts and not charge separately for individual guest VMs running on VM Hosts
45	Cover geographically distributed networks through multi-level scalable distributed deployment architecture
46	Ability to add new pollers at no extra cost.
47	Ability to collect data for this module from OT network through Unidirectional gateway (at physical level)

ITSM Tool

	General Requirement for ITSM
1	Low Code/No Code Platform for Integration:
	Tool should provide REST APIs to integrate with IT Infrastructure Management, Configuration Management, Network Management, CRM, SIEM tools to automate Events to Ticket.
2	The proposed tool must provide Dynamic REST/WEB API integration without writing any code, Just by configuration on GUI with various inbuilt Authentication method. Should be able to do 2 Way Integration with 3rd party and should be able to exchange the data at any stage of the lifecycle of the Event, Incident, Problem, Change, Request, Task, Release etc.
4	Team/Individual Pressure/Load Score, Efficiency Score, Expertise Score should be present on Dashboard
5	Tool must provide intelligent Email-to-Incident feature in which tool admin has the option to allow certain domains for automatic conversion of emails to tickets. Tool should merge all subsequent email communication for a particular email-to-incident ticket into the same ticket in the form of a message thread. Tool should be intelligent enough to understand email conversation chains for merging emails to a particular incident. Merging logic should be not only based on Ticket ID but also on email sender, cc responses to that email chain
6	Tool should be able to provide real-time Email, SMS Notification alerts to notify respective users about any changes in ticket state and status. Tool should provide Email Communication Interface to allow technicians to send replies to customers/ end users from the tool GUI and Record all the Email Communication in Chronological Order
7	The integrated ITSM module should have its own Android & IOS app
8	The proposed tool must provide GUI interface for users, requesters, customers, support staff, 3rd party vendors, Area Managers, Field Engineers, Site Engineers, Supervisors, Managers, (Helpdesk available. 24x7 at customer site) etc. with options to restrict amount of information that can be accessed by each role
9	The proposed tool should have the capability to be able to define and access the Business and Technical catalogue separately
10	The tool should allow the administrator to create categories and multiple sub- categories (under each category) in hierarchical order for services being offered to the end users.

11	Tool should have option to define the workflow for each service created in the Service Catalogue for each process (Incident, Problem, Change, Request)
12	The tool should provide a self-service portal where End-users/requesters can log in and raise incidents
13	End user should also be able to browse the service catalogue and should be able to raise request by providing the input to particular service request, Address etc.
14	The proposed tool must provide an option for users to track their incidents status, check knowledge articles and chat with technicians on particular incidents
15	Users must have an option to Log, Classify, Categorize & Prioritize Requests
16	The proposed tool must provide an option to design the request management input parameter template for each service in service catalogue
17	The proposed tool must provide an option to design dynamic workflows, lifecycle, Tasks, Notification action for each service which can be requested
18	The proposed tool must record Problem Record Date and Time, Problem Source, Contact Detail, Symptoms and Status
19	Tool should allow Problem records to be classified according to priority and category
20	Tool should allow a problem record to be escalated based on pre-established rules with option to manually overridden conditions
21	Tool should allow users to create, edit & delete Change Request through web interface, Each CR should have Unique and any authorized user should be able to raise RFC
22	The proposed tool must allow Change records Categorization, classification according to Change Class(Permanent, temporary, Recurring), Change Type (Normal, Standard, Expedited, Latent, Emergency), Change Category, service category, Impacted Service, Impact, Urgency, Priority.
23	The Change records in the proposed tool must contain State & Status information along with Dynamic Workflow definition with colour coding for each status
24	There should be records for Benefits, Effects, and Required resources & Cost required to perform change
25	The proposed tool must have a powerful knowledge management functionality
26	Knowledgebase Management should be integrated with the NMS/EMS system
27	The proposed tool must provide Role based, Team Based & User Based access control on KB articles/FAQ/Information/KE/Solutions etc.
28	In FAQ/Solutions type of knowledge, system should allow to add multiple questions/multiple solutions with single knowledge article
29	Tool should have option to promote knowledge to analysts (Service Desk) and end users (Service Portal)
30	Warranty & Support:- 5 years OEM warranty, 5-year parts, 5-year labour, 5-year onsite support.

Data Diode to restrict traffic from IT to OT system

Sl. No	Specifications
1	The device should provide secure electronic unidirectional data transfer between two IP-based LAN networks.

2	The data-diode should be of new generation, based on semiconductor technology, such as FPGA to ensure assurance of one-way communication.
3	The device should provide "Air-Gap" level of isolation while transferring data in a single direction.
4	The data-diode should operate at OSI Layer-1 and 2 only
5	The offered device should be a single unified device in a single 1U chassis, consisting of the Data Diode and related TX & RX units and without any need for additional hardware or require installation of any proprietary software on the interacting servers.
6	The data-diode should support high-speed data transfer with typ 1Gbps throughput.
7	The offered device shall go to FAIL-OPEN (disconnected) state in the event of power failure.
8	The device must guarantee to never allow full packet/session traversal through it. All packets/sessions must be terminated at the device.
9	The device should terminate all configured protocols and should be protocol aware to incoming traffic.
10	The device should maintain protocol integrity, facing the connected entities.
11	The offered device should use non-routable protocols to transfer data from source to destination server.
12	The device should support multiple protocols, as SFTP, HTTPS(PUT), SMTP(RELAY), SYSLOG (to be mentioned as per system need).
13	The offered device shall allow configuring customized ports for different services on both transmit and receive network segments, individually
14	The device offered should be able to run multiple TCP services simultaneously.
15	The SFTP server on the device should support upto 5 incoming SFTP clients and sessions with separate username and passwords, concurrently
16	The device should preserve file integrity and timestamp of files during data transfer.
17	The device should provide availability of details of data transfer transactions using Syslog service.
18	The device offered should provide information and alerts to external systems via Syslog service.
19	The device should be easy to implement using standard Ethernet ports on both sides.
20	The offered device should have an automatic clean-up mechanism on SFTP server side for non-configured file types and payload.
21	The device should provide near real-time data transfer with minimal latency.
22	Warranty & Support:- 5 years OEM warranty, 5-year parts, 5-year labour, 5-year onsite support.